CREATE Module: Perturbation Theory Lecture 1: Mechanical Models

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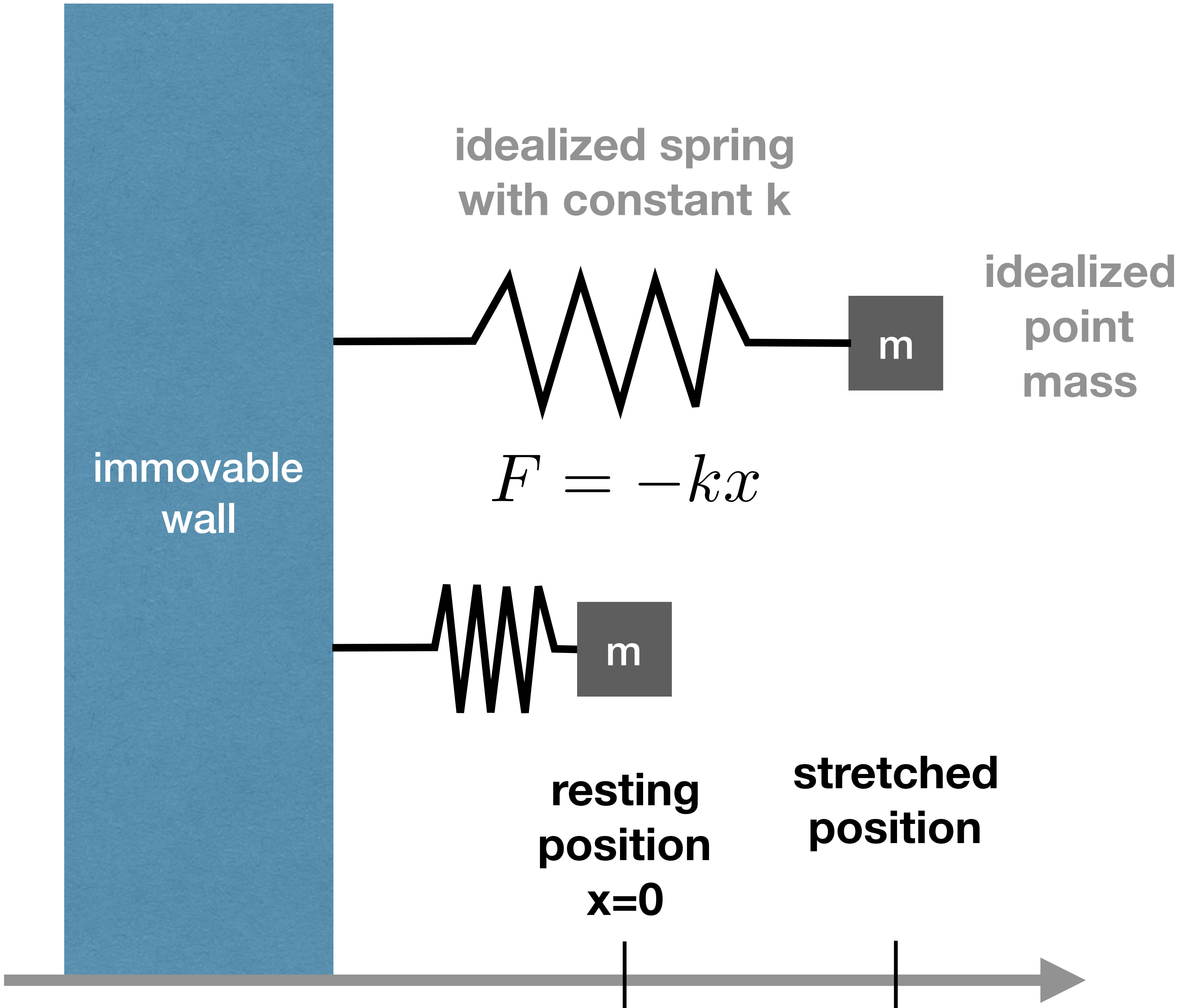
- The purpose of this lecture is three fold.
- First we introduce the mass on a spring model from classical mechanics in both linear and nonlinear form.
- Classical mechanics is simpler than fluid mechanics, and this model allows us to discuss certain aspects of model analysis with a set of well-known examples.
- Second we demonstrate some basics of analysis including non-dimensionalization and the derivation of an energy equation.
- Third we discuss the whole idea of mathematical modelling.

$$m\frac{d^2x}{dt^2} = -kx$$

Newton's Second Law



$$F = -kx$$

- The mass on a spring model, or SHO, assumes:
- Mass is conserved, and treated as a point mass.
- The spring force is linear and proportional to displacement.

$$v = \frac{dx}{dt}$$

$$m\frac{dv}{dt} + kx = 0$$

$$mv\frac{dv}{dt} + kvx = 0$$

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = mv\frac{dv}{dt} \qquad \frac{d}{dt}\left(\frac{1}{2}kx^2\right) = kxv$$

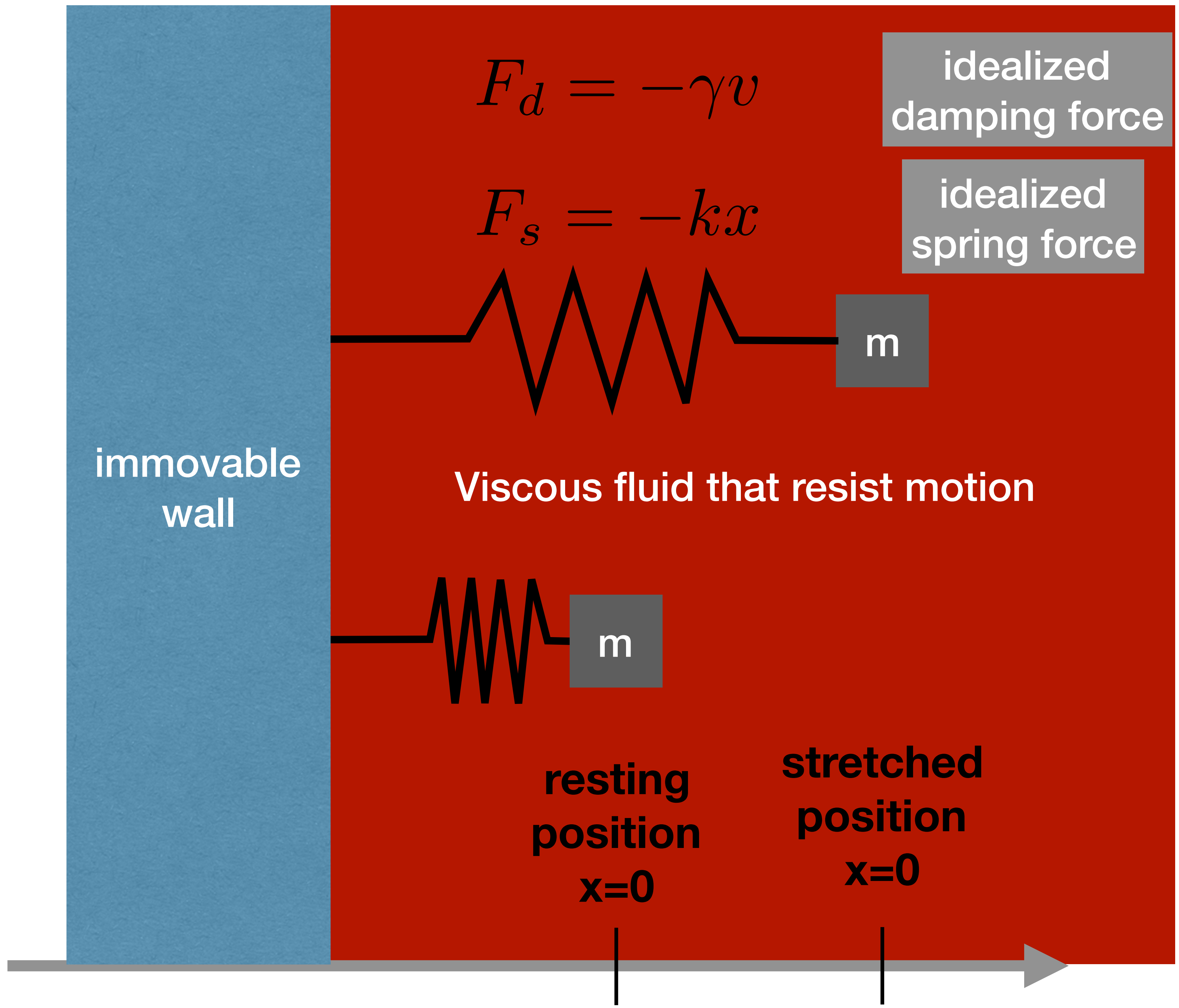
$$\frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2\right) = 0$$

$$E_k + U = \textbf{constant}$$

- The conservation of energy is an example of a first integral.
- This is a widely useful mathematical technique for both ODEs and PDEs.
- **One does not need a solution of the original DE.**

$$m\frac{d^2x}{dt^2} = -kx - \gamma\frac{dx}{dt}$$

Newton's Second Law



- The damped SHO keeps the assumptions of the SHO but adds:

- The damping is linear and proportional to velocity.

- Energy is defined as before and the first integral derivation follows the same mathematical pattern.

- But energy is no longer conserved.

$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = mv\frac{dv}{dt} \qquad \frac{d}{dt}\left(\frac{1}{2}kx^2\right) = kxv$$

$$\frac{d}{dt}\left(E_k + U\right) = -\gamma v^2$$

- Unless the mass is not moving, energy is always being taken out of the system.

- The rate of energy withdrawal is proportional to the kinetic energy and not the velocity.

- The SHO and the damped SHO are linear problems, and hence well known to have analytical solutions.

- The natural frequency of the SHO is usually written as follows:

$$\omega_0 = \sqrt{\frac{k}{m}}$$

$$x(t) = a\cos(\omega_0 t + \phi)$$

- a is the amplitude and phi the phase shift.

- The above is different than the ODE class way of writing the solution, but it is more common in physics/engineering references.

- Initial conditions for x and v are enough to specify all parameters uniquely.

- For the overdamped case D>0 and $m\lambda^2 + \gamma\lambda + k = 0 \implies \lambda = \dfrac{-\gamma + \sqrt{D}}{2m}$ has two real solutions.

- The definition $D = \gamma^2 - 4mk$ implies that both lambda values are negative.

- The general solution is thus given by $x(t) = c_1 \exp(\lambda_1 t) + c_2 \exp(\lambda_2 t)$ where the coefficients are determined by initial conditions.

- x(t) tends to zero monotonically and if x(0)>0, x(t) never cross zero.

- The damped SHO is solved by assuming a trial solution: $x(t) \propto e^{\lambda t} = \exp(\lambda t)$

If any of this is new, it makes sense to add to the lab book as a "review" problem

- Taking the derivatives of the exponential to bring down a lambda each time and cancelling the common factor of the exponential gives a quadratic in lambda

- $m\lambda^2 + \gamma\lambda + k = 0 \implies \lambda = \dfrac{-\gamma + \sqrt{D}}{2m}$

- Where $D = \gamma^2 - 4mk$ is the discriminant, the sign of which sets the type of solution found.

- The three types of solutions are called: overdamped (D>0), critically damped (D=0), and underdamped (D<0).

- For the underdamped case D<0 so that $\sqrt{D} = i\sqrt{-D}$.
- This means lambda is complex:

$$\lambda = -\frac{\gamma}{2m} \pm i\frac{\sqrt{4mk - \gamma^2}}{2m}$$

- The solution of the second order ODE is built from complex exponentials. Check standard textbooks for the algebra, but the answer is: $x(t) = a\exp(-\gamma t/2m)\cos(\omega_d t + \phi)$ where the frequency is given by the imaginary part of lambda.

- I leave the critically damped case as an exercise for you.

- We will generally focus on the underdamped case in this course:

$$\omega_d = \frac{\sqrt{4mk - \gamma^2}}{2m}$$

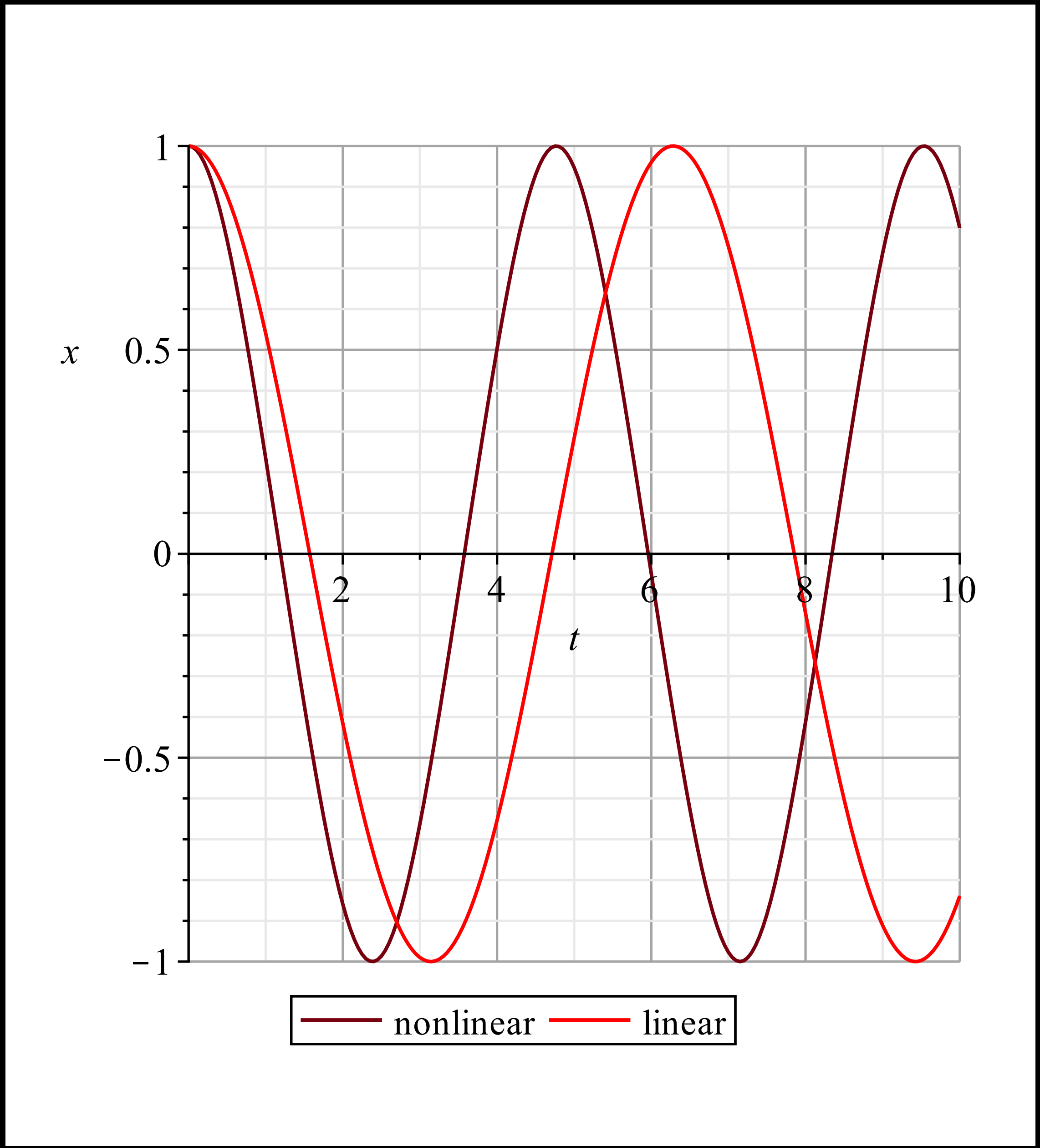
$$x(t) = a\exp(-\gamma t/2m)\cos(\omega_d t + \phi)$$

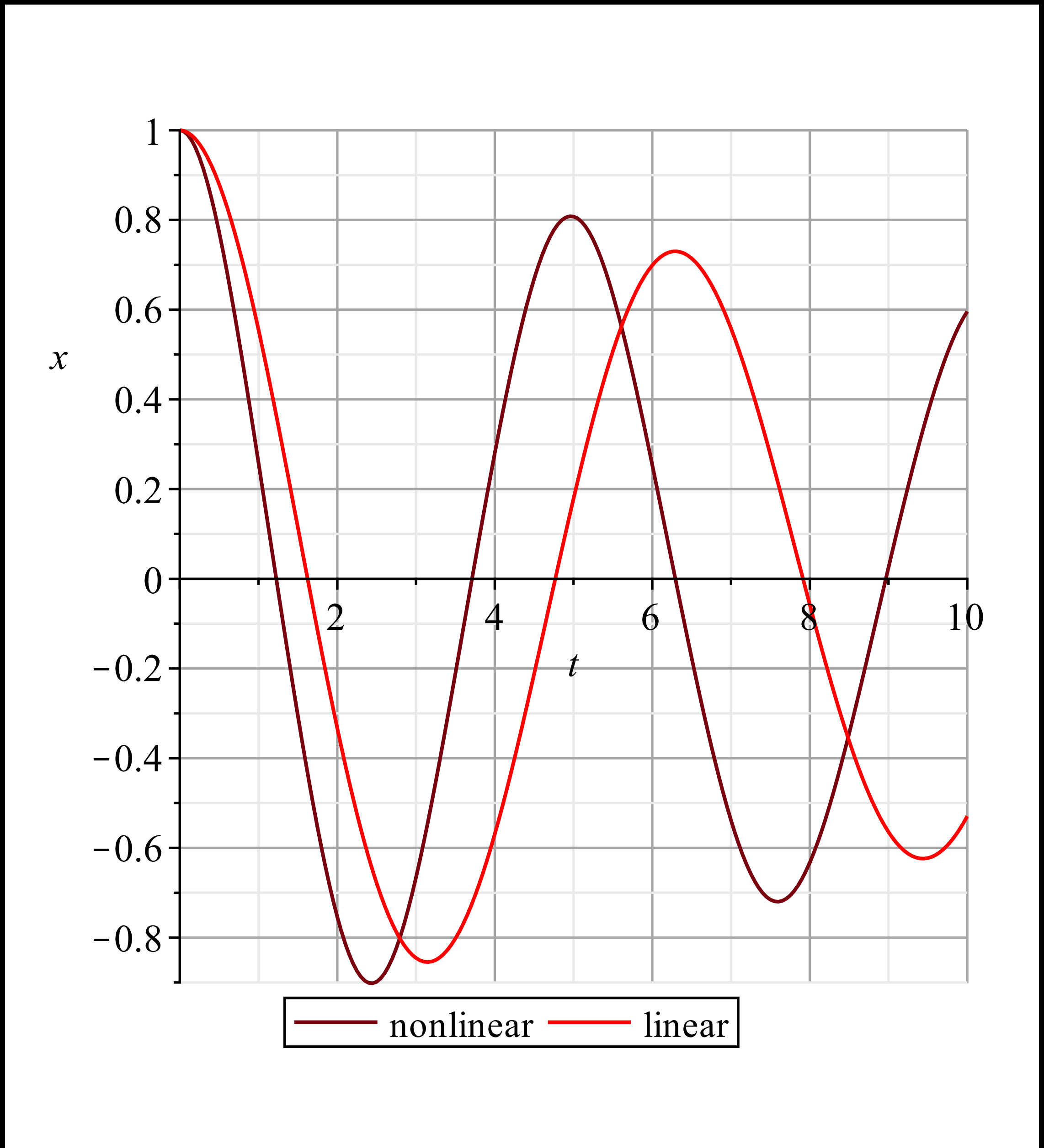
- Again a is the amplitude and phi the phase shift.

- The decay rate depends only on the damping parameter and the mass.

- The frequency of oscillation depends on all the physical parameters.

- Sample solutions of the SHO (left) and damped SHO (right).





- Focus on the red curves for now.

- Nonlinearity can enter the SHO in modeling the force or the damping.

- The traditional way to think of this is as the linear spring force with a small correction (often called the Duffing oscillator).

- **To maintain a force that is restoring only odd terms can appear.**

- Energy is still conserved but the mathematical form of U is different.

$$v = \frac{dx}{dt}$$

$$m\frac{dv}{dt} + kx + k_3x^3 = 0$$

$$mv\frac{dv}{dt} + kvx + k_3vx^3 = 0$$

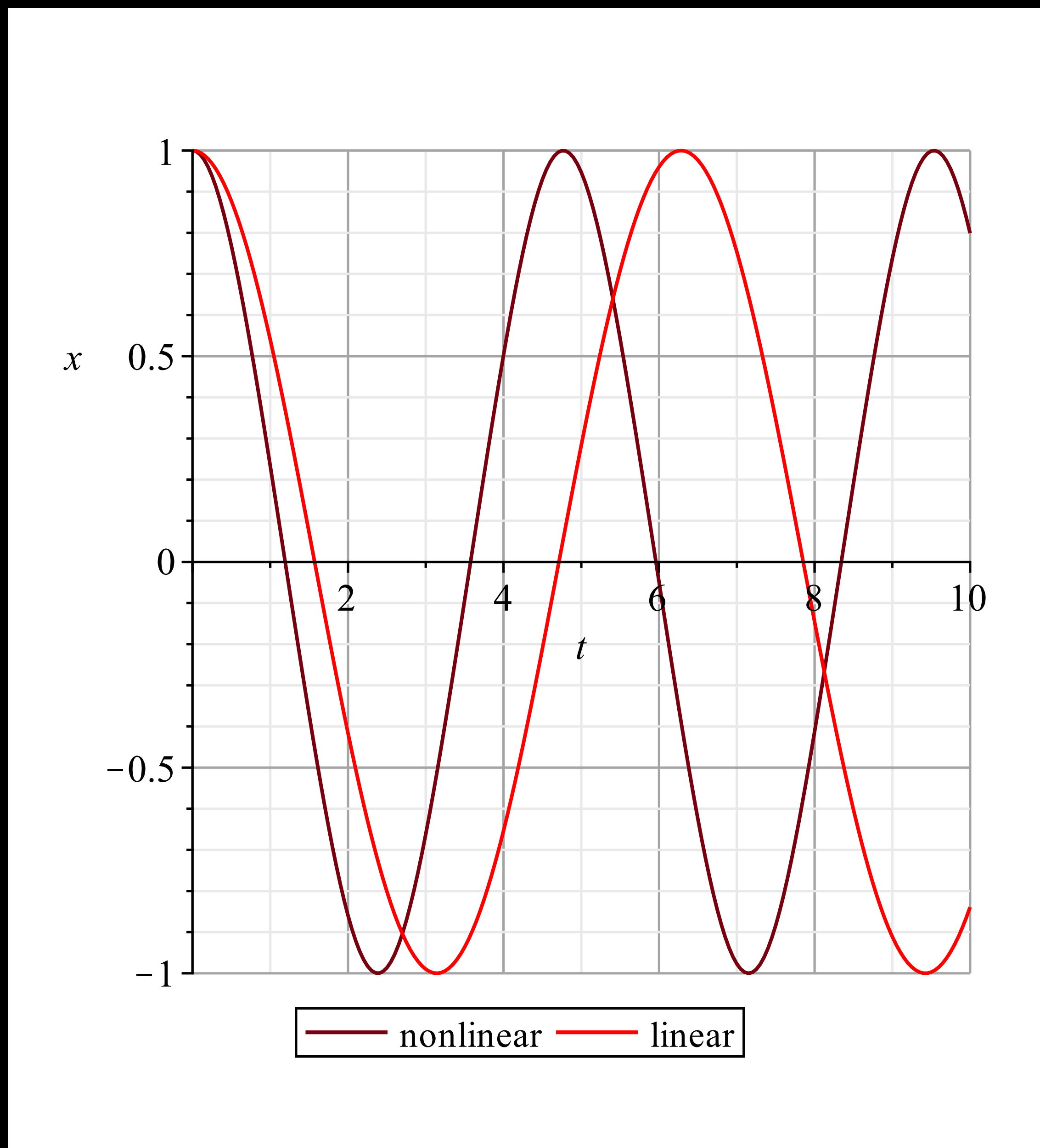
$$\frac{d}{dt}\left(\frac{1}{2}mv^2\right) = mv\frac{dv}{dt}$$

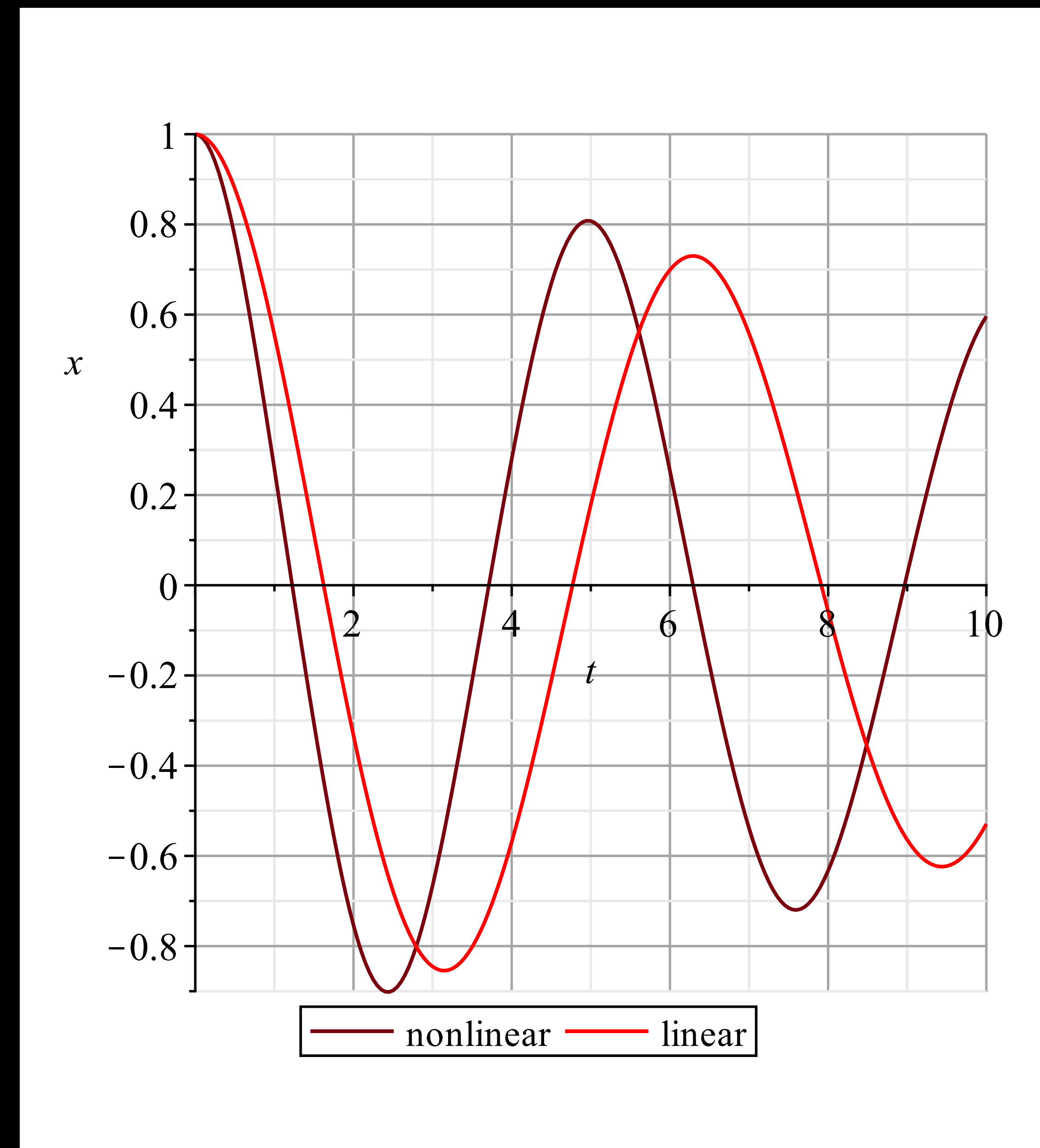
$$\frac{d}{dt}\left(\frac{1}{2}kx^2 + \frac{1}{4}k_3x^4\right) = kxv + k_3vx^3$$

$$\frac{d}{dt}\left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 + \frac{1}{4}k_3x^4\right) = 0$$

$$E_k + U = \textbf{constant}$$

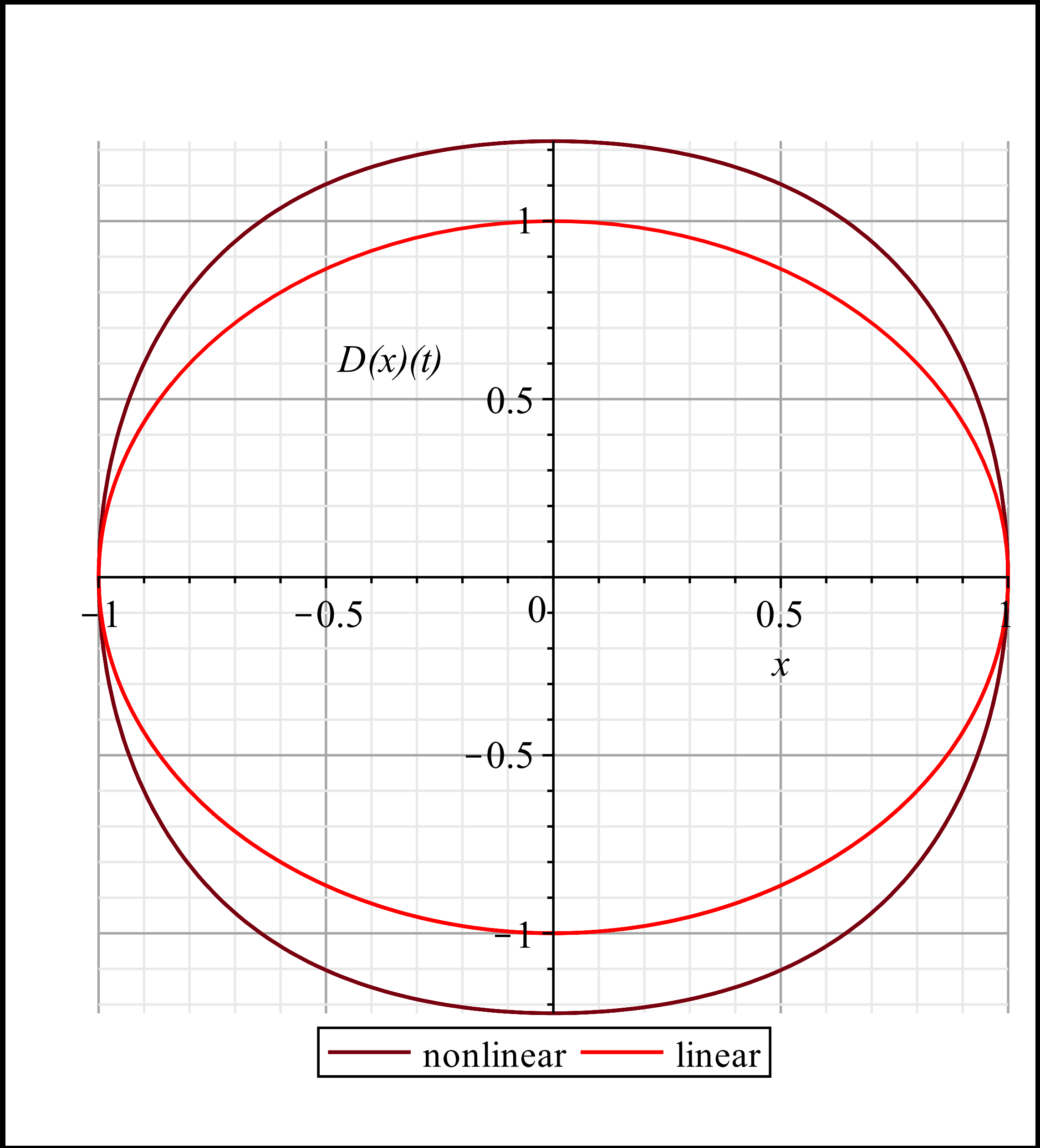
- Black curves show the nonlinear solution

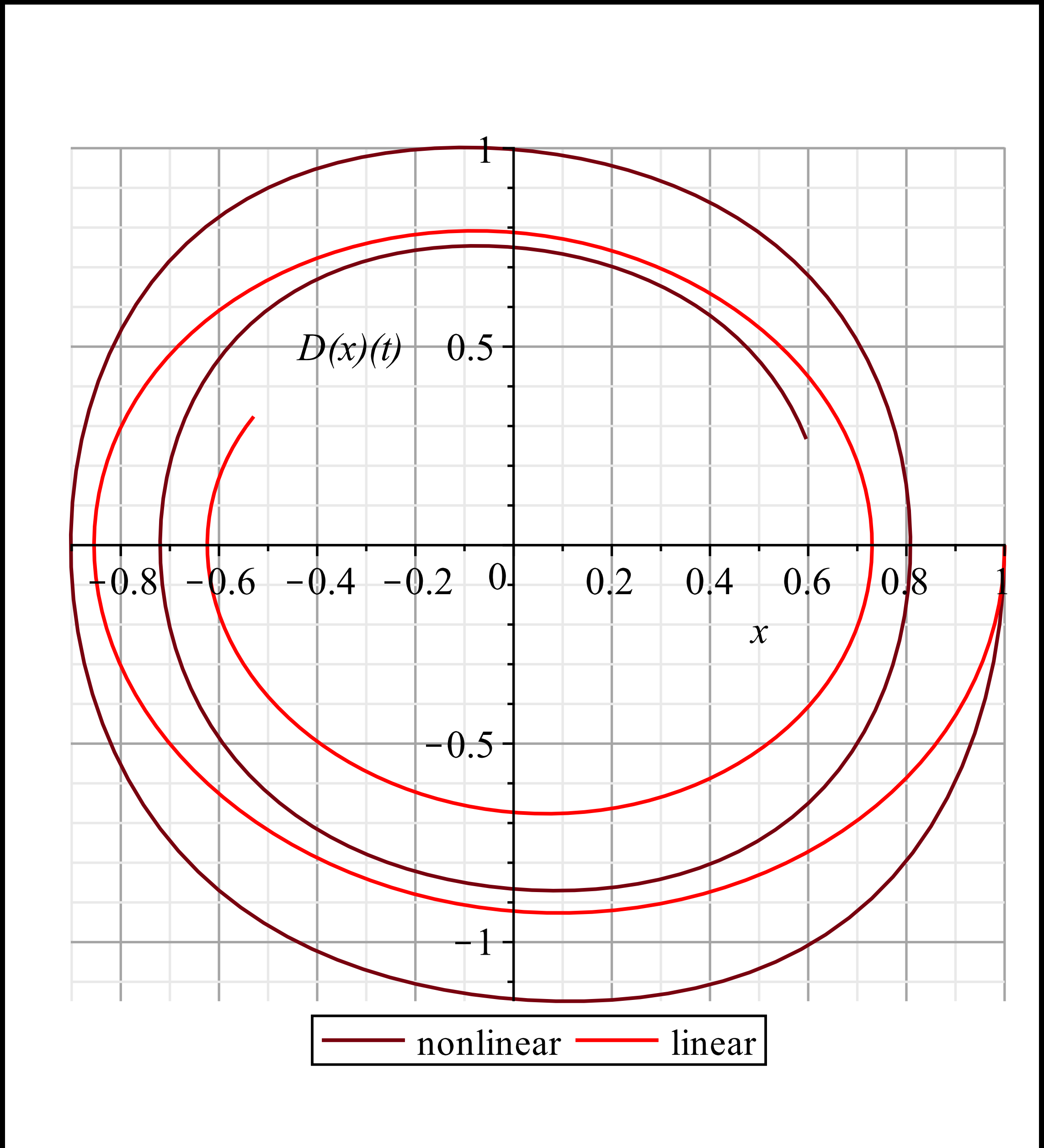




- Qualitative behaviour is similar, with only details differing.

- Phase portraits allow use to pinpoint the effects of nonlinearity





- The conservative case shows a departure from an ellipse.

- Let’s say we had a model but couldn’t decide if we really thought all the terms were important.
- How could we decide in a mathematically unambiguous way?
- Non-dimensionalization provides an answer.

$x = Ly$ **,** $t = Ts$ **where** y, s **are dimensionless**

$$\frac{mL}{T^2}\frac{d^2y}{ds^2} + \frac{\gamma L}{T}\frac{dy}{ds} + kLy = 0$$

$$\frac{d^2y}{ds^2} + a\frac{dy}{ds} + y = 0 \text{ **where** } a = \gamma T$$

- But how to choose T? And why does L not matter?

- Answering why L does not matter is easy. Basically, for any linear equation the amplitude parameter can be anything, and here L is the typical scale of x(t).

- Choosing T is harder and involves implicit assumptions.

- This can be a bit scary, but if you think about it, math all by itself cannot answer applied problems and at some point you need to take a position.

- One easy choice is to say, “I think that damping is fairly weak so that I basically have a small change to the undamped SHO”.

- For the undamped SHO the natural frequency provides a time scale, and we can just choose that, namely:

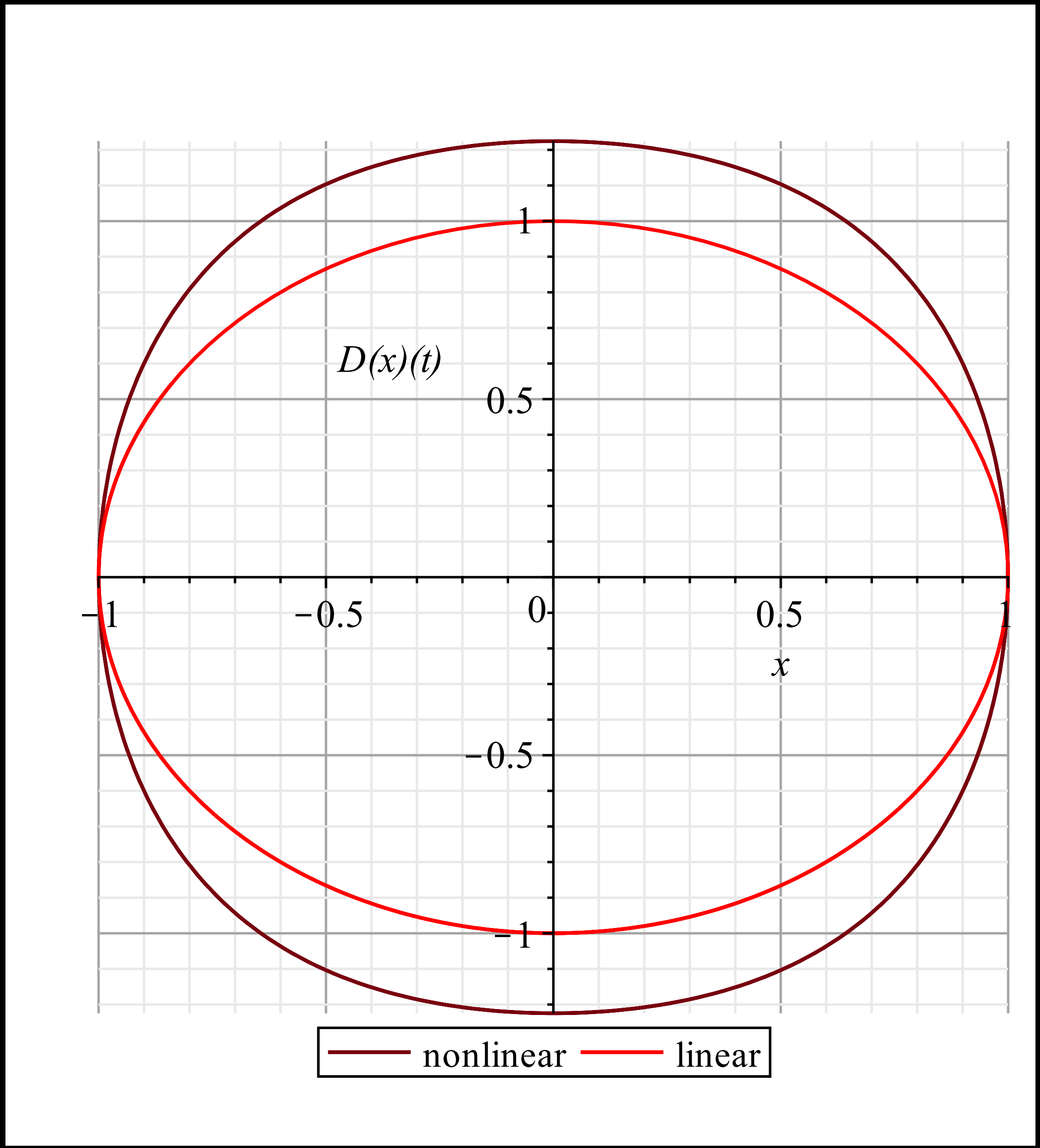
$$T = \frac{1}{\omega_0} = \sqrt{\frac{m}{k}}$$

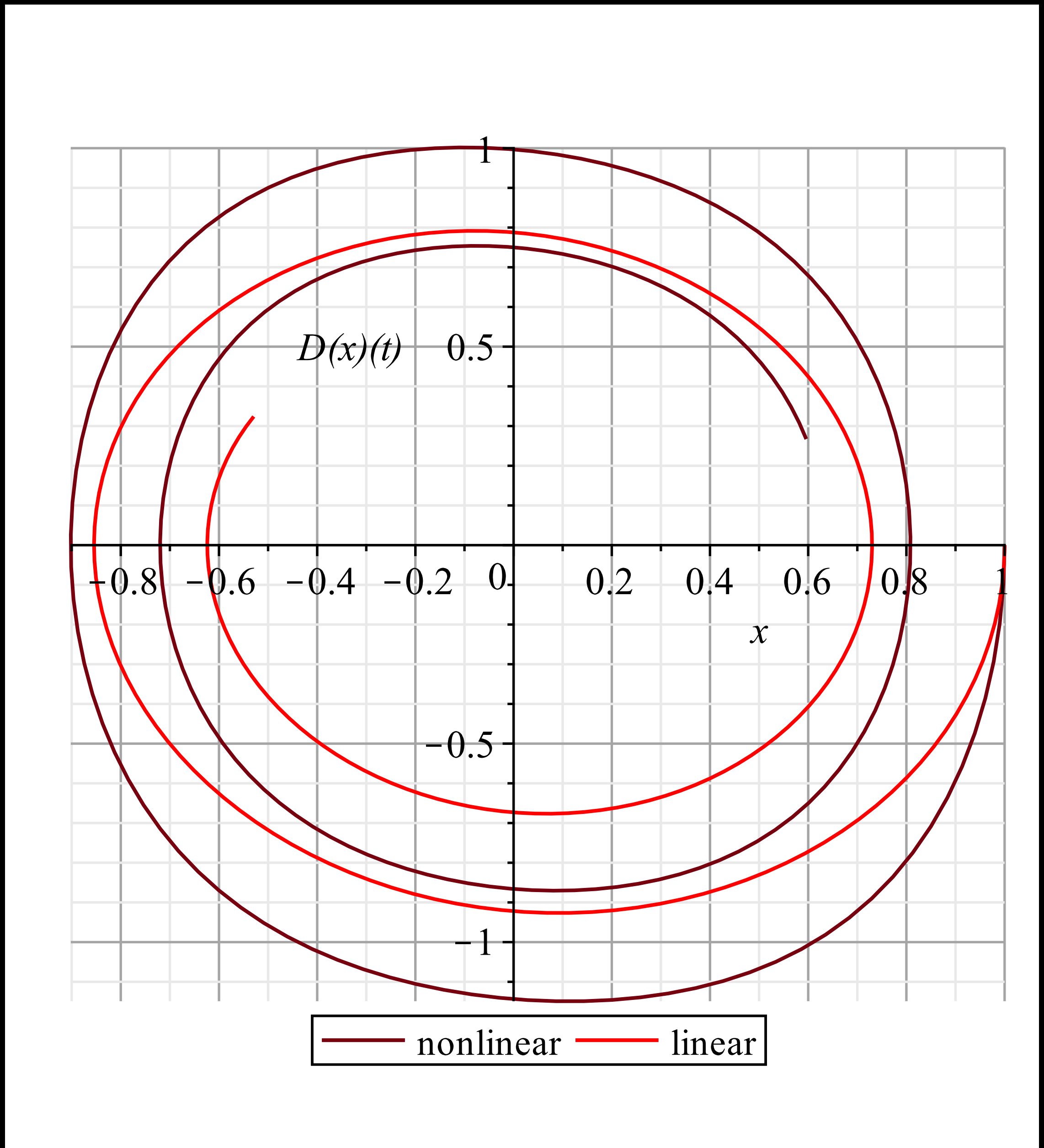
$$\frac{d^2y}{ds^2} + a\frac{dy}{ds} + y = 0 \text{ where } a = \gamma T$$

$a \ll 1$ **when** T **fixed,** $\gamma \ll 1$, **or** γ **fixed,** $T \ll 1$

- So what does the dimensionless form get you?
- Well, first of all, we can unambiguously classify the solutions.
- The discriminant is: $a^2 - 4$
- This means if a<2 we have an underdamped system.
- We could also imagine doing some sort of “weak” damping approximation: $a \ll 1$
- This, as it turns out, will take us some time to puzzle out.

- For the nonlinear case (black) L will not cancel out.





- Let's consider the undamped Duffing oscillator.

$$m\frac{dv}{dt} + kx + k_3x^3 = 0$$

- Let’s consider a Duffing oscillator for which we implicitly assume that the nonlinear effects are secondary to the primary oscillation, but not negligible.

- We again seek an unambiguous way to answer the question: “How important is the nonlinearity?”

$x = Ly$ **,** $t = Ts$ **where** y, s **are dimensionless**

$$\frac{mL}{T^2}\frac{d^2y}{ds^2} + kLy + k_3L^3y^3 = 0$$

Let $T = \dfrac{1}{\omega_0} = \sqrt{\dfrac{m}{k}}$

$$k\frac{d^2y}{ds^2} + ky + k_3L^2y^3 = 0$$

$$\frac{d^2y}{ds^2} + y + Ay^3 = 0 \text{ **where** } A = \frac{L^2k_3}{k}$$

- How to choose L?

$$\frac{d^2y}{ds^2} + y + Ay^3 = 0 \text{ **where** } A = \frac{L^2k_3}{k}$$

- If we specify the initial position x0 and set the initial velocity to be zero, which is a common thing to do in oscillator problems then L=x0 is a good choice.

- If we have a non-zero velocity v0 things are a little more complicated but we can use the conservation of energy to equate the initial energy to the energy when the velocity is zero.

- This will give the furthest deviation from the resting position, and an unambiguous way to set L.

- The equation to solve is quartic, but only even powers appear so this is quite doable.

$$\frac{1}{2}mv_0^2 + \frac{1}{2}kx_0^2 + \frac{1}{4}k_3x_0^4 = \frac{1}{2}kL^2 + \frac{1}{4}k_3L^4$$

- In summary, linear equations may be non-dimensionalized, or scaled out in a way that is amplitude independent.
- The relative strength of various terms can then be compared unambiguously.
- For nonlinear equations the amplitude is expected to play a role in the non-dimensionalization.
- For oscillators, the expression for total energy is a good way to estimate an amplitude from the initial conditions.

Mathematical Modelling

- Classical mechanics and mathematics have such an intertwined history the two almost seem to be the same thing.

- However, mathematical modelling, of which the classical mechanics models I have presented are an example, is its own discipline.

- **The idea is not to reproduce the world, with a mathematical version.**

- Instead the idea is to create a mathematical representation of some essential aspect of a phenomenon.

- Mathematical models thus need a rubric for evaluating how useful they are.

- A very simple example is a model to predict the NYSE stock index.

- Such a model would probably be evaluated in a pretty straight forward way, since we can track the NYSE and the model for some period of time (say a month) and construct an actual error (probably the absolute value).

- Alternatively we could see if the model makes us money or loses us money. This latter test would more or less ignore how we got to the final state.

- Also, because stocks don't have a Newton's law, there is nothing saying that the model should take the mathematical form of a differential equation.

- Fluid mechanics is an older science and does have its fundamental governing equations:

- $\rho_o \dfrac{D\vec{u}}{Dt} = -\nabla p + \mu \nabla \vec{u}$ is the Newton's Law. To see how recall that the stress-rate of strain relation is $\tau = -p\mathbf{I} + 2\mu\mathbf{e}$ so that the right hand side can be written as $\rho_o \dfrac{D\vec{u}}{Dt} = \nabla \cdot \tau$ and now the LHS is the "ma" bit and the right side is the force.

- Because fluid mechanics is a "PDE theory" the way perturbation theory shows up is subtle, and a bit ad hoc. That's why we will use simpler models for the first four lectures.